

FAA, MIKE MONRONEY AERONAUTICAL CENTER

Statement of Work

Environmental Architectural/Engineering Services

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DIVISION 0 GENERAL REQUIREMENTS

SECTION 010 Scope of Work

1. The Federal Aviation Administration's (FAA) Mike Monroney Aeronautical Center (MMAC), located in Oklahoma City, Oklahoma, is the site of a variety of activities supporting the central mission of the FAA, which is to promote safety and efficiency in the use of the nation's airspace. Environmental Architectural and Engineering (AE) services are needed to ensure compliance with local, state and federal environmental laws in the conduct of the FAA's mission.
2. Services to be provided under this contract are defined as Type A and Type B as follows:
 - 2.1. Type A: Services will provide field surveys and investigations required to obtain preliminary design data and other services. However, it is understood that MMAC reserves the right to utilize the services of other consulting firms or to directly contract environmental remediation work at MMAC's discretion. Work under Type A includes, but is not limited to:
 - 2.1.1. Development of program requirements;
 - 2.1.2. Determination of project feasibility;
 - 2.1.3. Preparation of drawings of an existing or new facility, where current drawings are not available;
 - 2.1.4. Subsurface investigations (soil borings);
 - 2.1.5. Architectural, structural, electrical, and mechanical investigations of an existing building, where current information is not available;
 - 2.1.6. Surveys (topographic, boundary, utility);
 - 2.1.7. Preparation of models, color renderings, photographs, or other presentation materials;
 - 2.1.8. Travel and per diem associated with directed trips exceeding 25 miles from contractor's facility;
 - 2.1.9. Supervision and/or inspection of construction;
 - 2.1.10. Preparation of operating and maintenance manuals;
 - 2.1.11. Budget and cost estimates;
 - 2.1.12. Submittal review;
 - 2.1.13. Preparation of electronic media;
 - 2.1.14. Any topographical surveys, excavations, soil borings, soil analyses, chemical tests, laboratory tests, and any other tests required for a complete design of each project will be the responsibility of the AE.
 - 2.1.15. Preparation of contract as-built drawings.
 - 2.2. Typical areas where Type A studies may be required include:

- 2.2.1. Spill prevention, containment and countermeasure plans.
- 2.2.2. Noise evaluations.
- 2.2.3. Industrial wastewater pretreatment.
- 2.2.4. Air quality, including but not limited to: air emission inventories per the Environmental Protection Agency (EPA) and Oklahoma Department of Environmental Quality (ODEQ), indoor air quality assessments involving vapor intrusion to the indoor environment and human health risk assessment from an inhalation standpoint per EPA/ODEQ guidance.
- 2.2.5. Industrial waste hauling, control and treatment.
- 2.2.6. Asbestos survey, testing and abatement.
- 2.2.7. Environmental Impact Statements (EIS's) and Environmental Assessments (EA's) under the National Environmental Policy Act (NEPA).
- 2.2.8. Environmental Due Diligence Audits (EDDA's) and Environmental Site Assessments (ESA's) under the Comprehensive Environmental Response, Compensation and Liability Act (CERCLA).
- 2.2.9. Hazardous material storage and handling.
- 2.2.10. Groundwater assessments, monitoring well installation, well plugging and abandonment.
- 2.2.11. Superfund Amendments and Reauthorization Act (SARA) Title III program.
- 2.2.12. Polychlorinated Biphenyl (PCB) survey, testing and abatement.
- 2.2.13. Solid waste management.
- 2.2.14. Hazardous waste management.
- 2.2.15. Pollution prevention opportunity assessment.
- 2.2.16. Fuel storage tank management.
- 2.2.17. Environmental compliance audits.
- 2.2.18. Environmental management system audits.
- 2.2.19. Remediation of air, soil, sediment, groundwater and/or surface water contamination, using Green and Sustainable Remediation (GSR) where applicable.
- 2.2.20. Drinking water quality analysis and treatment.
- 2.2.21. Environmental contaminants of emerging concern (CEC's); including but not limited to per-and polyfluoroalkyl substances (PFAS).
- 2.2.22. Natural and cultural resource surveys, such as protected species surveys, wetland surveys, etc.

2.2.23. Impaired waters / streams and watershed studies under the Clean Water Act (CWA).

2.2.24. Vapor Intrusion surveys, investigations and mitigation.

2.3. Type B - Services will provide construction documents for competitive bidding and associated construction administration. Work under Type B will include but not be limited to the following:

2.3.1. Field visits;

2.3.2. Preparation of construction drawings;

2.3.3. Preparation of construction specifications;

2.3.4. Construction cost estimates;

2.3.5. Design analysis;

2.3.6. Economic analysis; and

2.3.7. Preparation of project documentation in approved electronic media format.

2.3.8. Review of construction documents by local city and state code enforcement officials.

3. The AE will be responsible for the verification of all dimensions, measurements, and location of existing facilities, utilities, and equipment. The AE will make visits to the project site and to the Resident Engineer's (RE) office as necessary for effective completion of each project's design. The AE is to do sufficient research to provide a design in compliance with each project's scope.
4. The contractor will furnish all labor, tools, materials, equipment, supervision, personnel and services as required to complete work as described in this specification.
5. All FAA requirements under this contract will be on a task order basis as requested by the government. The FAA will prepare a scope of work for each task to be performed. The contractor will then provide a negotiable proposal and price quotation based on the given scope of work. The proposal will include a description of work (work plan), manpower and equipment requirements and a list of deliverables. In addition, the proposal will include a work schedule and will address the preparation of amended health and safety plans, field sampling plan and other such plans as appropriate. The price quotation will conform to the listed fixed rates for hourly rates, direct cost, equipment and testing and will include an analysis of annual operating and maintenance costs, if applicable. Specific delivery order performance periods, deliverable item requirements, and delivery dates for all delivery orders placed pursuant to this contract will be established at the time any such delivery order is issued.
6. The contractor will give the fullest consideration practicable to the availability of resources and personnel required and the establishment of priorities of work. Work under such additional task orders will not commence until the complete task order is finalized and signed by the Contracting Officer (CO). Receipt of a signed task order will constitute a notice to proceed (NTP) unless a different NTP is specified in the task order.

SECTION 020 Facility Background Information

1. Activities and Mission:

- 1.1. The Federal Aviation Administration's (FAA) Mike Monroney Aeronautical Center (MMAC), located in Oklahoma City, Oklahoma, is the site of a variety of activities supporting the central mission of the FAA, which is to promote safety and efficiency in the use of the nation's airspace. In fulfillment of this mission, the MMAC conducts centralized training, aircraft fleet maintenance and modification, central warehousing and supply, aeromedical research, and maintains and administers aircraft and airman (including medical) records. It consists of approximately 1100 acres of land and numerous buildings which are leased in total from the Oklahoma City Airport Trust (OCAT).
- 1.2. Waste Generation History - The primary MMAC activities that use and generate hazardous materials/hazardous wastes include aircraft maintenance; facilities and vehicle maintenance; painting and plating; material fabrication and repair; fuel storage, transfer and distribution; and aeromedical research. Most hazardous waste generated by FAA have been containerized and disposed by contractors off-site. Past disposal practices on the MMAC lease included burial of radioactive materials, discharge to sanitary and storm sewers, and on site burning by the former tenant prior lease by the FAA in 1958. These practices, along with their groundwater impacts, have been the subject of a comprehensive environmental investigation since 1992. Past and present sites investigated, and the current status, is shown in Attachment A.

SECTION 030 Site Conditions

1. Location:

- 1.1. The MMAC is located in the far southwest portion of Oklahoma County and is also in the southwestern region of the Oklahoma City Metropolitan Area. The facility is within Oklahoma City limits. Cities to the south of the facility are located in Cleveland County and include the cities of Moore and Norman.

2. Climate:

- 2.1. The climate of Oklahoma is continental in type, as in all of the central great plains. Summers are long and occasionally very hot. Winters are shorter and less rigorous than those of the more northern plain states.
- 2.2. The climate of Oklahoma City falls mainly under the continental controls of the great plain's region. During the year, temperatures of 100 degrees or more occur on an average of 10 days but have occurred on as many as 50 days or more. While summers are usually hot, the discomforting effect of extreme heat is considerably mitigated by low humidity and the prevalence of a moderate southerly breeze. The average date of the last freeze is early April, and the average date of first freeze is early November. Freezes have occurred in early October.
- 2.3. Summer rainfall comes mainly from showers and thunderstorms. Winter precipitation is generally associated with frontal passages. The period with the least number of days with precipitation is November through January and the month with the most rainy days is May. Thunderstorms occur most often in late spring and early summer. Large hail and/or destructive winds on occasion accompany these thunderstorms. Snowfall averages less than 10 inches per year and seldom remains on the ground very long. Occasional brief periods of freezing rain and sleet storms occur. Heavy fog is infrequent. Prevailing winds are southerly except in January and February when northerly breezes predominate.

3. Environment:

- 3.1. MMAC is located on generally flat, grassy, prairie-type terrain at an elevation of 1265 to 1280 feet. The Center is situated on a ridge dividing the North Canadian River basin to the north and the Canadian River basin, to the south. Tree-lined drainages extend from the Center to the south and east into branches of Cow Creek. The MMAC lease surrounds an Outlet Lagoon which receives stormwater drainage from the Center and the Will Rogers International Airport. The Lagoon drains into Lake Peachy, headwater for Cow Creek. Although some residences are within a mile of the Center, the area is generally rural, with businesses to the north and west. Will Rogers International Airport is located on the East, sharing part of the flight line with the FAA. Oil wells operate on site.

SECTION 040 General Information

1. Definitions:

- 1.1. The abbreviation CO/COR appearing herein are an abbreviation for Contracting Officer/Contracting Officer's Representative.
- 1.2. The term contractor refers to the prime contractor, partners in a joint venture and any subcontractors used by the primary contractor.
- 1.3. The RE refers to the MMAC resident engineer assigned as point of contact for a particular task.

2. Inspection of AE's Facilities:

- 2.1. The CO or his authorized representative may at all reasonable times, inspect or otherwise evaluate the work being performed here under and the premises on which it is being performed. If any inspection or evaluation is made on the premises of the AE or his subcontractor, the AE will provide and will require his subcontractors to provide all reasonable facilities and assistance for the safety and convenience of the Government representatives in the performance of their duties. All inspections and evaluations will be performed in such a manner as will not unduly delay the work.

3. Commencement of Performance:

- 3.1. The execution of a contract will not constitute authority for the contractor to commence performance. Performance will be ordered by the issuance of a formal delivery order by an authorized CO of the Mike Monroney Aeronautical Center. Orders issued orally or by written telecommunications will reference the formal delivery order number and will be confirmed by issuance of the formal delivery order.

4. Inspection of Site:

- 4.1. Before submitting proposals for delivery orders, the contractor will visit the site to satisfy himself as to the nature and scope of all work to be done. The submission of a proposal will be taken as evidence that such an examination has been carried out and difficulties, if any, noted. Later claims for labor, work, material and equipment required for any difficulties encountered which could have been foreseen, will not be recognized and all such difficulties will be properly taken care of by the contractor at no additional expense to the Government.

5. Coordination:

- 5.1. General: The work of this contract includes coordination of the entire work of the project, including preparation of general coordination drawings/diagrams/schedules, and control of site utilization; from the beginning of activity, through the project close-out and warranty periods.
- 5.2. The contractor will meet with the COR prior to the start of any construction and will arrange necessary access and delivery routes between public thoroughfares and the construction site, and necessary storage and staging areas.
- 5.3. The contractor will schedule his work so that his working hours and the working hours of the Government's facility coincide. Should it be necessary to work hours other than normal, regular working hours, the contractor will arrange for access to the area with the COR at the site three working days prior to this need. Normal working hours are between the hours of 8:00 a.m. and 4:30 p.m. Monday through Friday, excluding Government holidays.

- 5.4. The contractor will keep on the construction site a set of drawings so that the exact location of all underground utilities and all equipment will be indicated thereon and turned over to the COR upon completion of the project.
- 5.5. The contractor will provide sufficient safe and proper facilities for the inspection of the work by the CO/COR. or his authorized representative.
- 5.6. The contractor will properly protect all his work from damage and will protect the Government's property from injury or loss during the performance of this contract. He will make good any damage, injury or loss, except such as may be directly due to errors in the contract documents or caused by agents or employees of the Government. He will adequately protect adjacent property as provided by law and the contract documents. He will provide and maintain all passageways, guard fences, lights and other facilities for protection required by public authority or local conditions.
- 5.7. The contractor will be responsible for damages to utility infrastructure, the work of other contractors or to buildings, or their contents, people, etc., caused by any of the equipment installed by him through equipment or material failures or disconnected pipes, fittings or by over-flows and will make, at his own expense, all repairs to merchandise, fixtures and equipment so damaged.
6. Temporary Facilities:
- 6.1. Temporary facilities may include those necessary for loading drums, decontamination areas, dressing rooms, equipment storage and on-site food, lodging or sanitation structures or modifications to structures.
- 6.2. The contractor will develop a decontamination plan as a portion of the Safety Plan and identify any temporary decontamination facilities to be constructed or used for the activities in this contract. One or more decontamination stations may be established. The decontamination facility (Contamination Reduction Zone) should be in an area between the Exclusion Zone (the contaminated area) and the Support Zone (the clean area). All equipment, structures and facilities used for decontamination must be decontaminated and/or disposed of properly. Contaminated equipment, temporary structures and materials should be dismantled or collected and placed in containers and labeled. Also, all spent solutions and wash water should be collected and disposed of properly. Any soils which have been contaminated should also be removed and placed in containers for testing. Any structures or equipment which is contaminated during the operation should be cleaned to remove any chemical contaminants. The decontamination stations will be removed or dismantled once they are no longer needed.
- 6.3. Any temporary structure or facility constructed for staging or loading drums or removal of soil will be designed to protect the environment from chemical contamination. Once operations are completed, the structure or facility will be removed and disposed of according to the amount and type of chemical contamination and applicable regulations. Any ground materials which were contaminated during staging or loading drums will be excavated and the material placed in containers for sampling. Dressing areas or equipment storage facilities constructed or used during site operations will be removed or returned to their original condition once they are no longer needed. Facilities and structures used by the contractor will be returned to their original condition, and any improvements will remain as directed by the CO or COR
- 6.4. Any modifications to FAA structure, buildings or facilities will be approved by the contracting officer prior to modification. Once the structures or buildings are no longer needed, they may be removed and/or the facility restored to its original condition. The CO has authority to approve any modification and to require removal or alteration of modifications once a building or structure is no longer needed.

- 6.5. Should the contractor require any power or utility shutdown during construction, he will arrange for such shutdown with the COR ten working days prior to this need.
- 6.6. Temporary utilities (water and electricity) will be furnished to the contractor by the Government except where remoteness of a site work location precludes this, in which case, the contractor will furnish its own water and electricity. Contractor will maintain and protect all utilities furnished to it by the government during construction and will repair or replace any items damaged through his own negligence.
- 6.7. Temporary toilet facilities will be furnished by the contractor. Facilities will be maintained in a clean and sanitary condition throughout the project period. At the discretion of the CO/COR the contractor may apply for and be granted permission to utilize the government's facilities at no cost to the contractor.
- 6.8. The contractor will provide temporary fences and barricades as needed for protection and security during construction.
7. Mobilization and Site Preparation:
- 7.1. The contractor will acknowledge that a careful investigation has been made of the nature and location of the work to be performed; the character, quality, and quantity of materials and obstructions to be encountered; the character of equipment and facilities needed prior to and during execution of the work; the general and local conditions and all other such matters which can in any way affect the work.
- 7.2. Mobilization - The contractor will:
- 7.2.1. Purchase and transport to the site appropriate containers for holding water, soil or other contaminated material generated during sampling procedures and 55-gallon drums suitable for storage and shipment.
 - 7.2.2. Prepare the necessary plans and procedures (see Division 1).
 - 7.2.3. Arrange for use or provide all personnel and equipment necessary to: (a) place soil, sampling debris, etc., in containers, (b) move containers into a storage/containment area, (c) label and mark containers (d) sample air, soil, ground water, surface water and sediment, (e) and decontaminate equipment and personnel.
 - 7.2.4. Arrange for food and lodging for workers during the duration of on-site activity.
- 7.3. Site Preparation
- 7.3.1. The contractor will prepare the site for sampling activities to ensure sampling goes smoothly and that worker safety is not compromised. Safety measures during site preparation should be afforded the same level of care as during actual sampling.
 - 7.3.2. The contractor will, as needed: (a) arrange traffic flow patterns to ensure safe and efficient operations and minimize inconvenience to MMAC employees and contractors, (b) eliminate physical hazards from the work area as much as possible, (c) construct operations pads for mobile facilities and temporary structures, processing and staging areas and decontamination pads, (d) provide adequate illumination for work activities, and (e) install all wiring and electrical equipment in accordance with the National Electrical Code.

8. Demobilization and Site Cleanup:

- 8.1. The contractor will place in containers and remove any materials contaminated with toxic/hazardous chemicals released because of his on-site operations. Contaminated materials will be cleaned up or removed immediately following the release.
- 8.2. All materials contaminated with toxic/hazardous chemicals will be placed in drums for storage and subsequent disposal. The contractor will ensure only those materials which have been contaminated will be placed in drums. Small amounts of hazardous waste such as may be generated during sampling will be containerized, labeled and prepared for shipment. At the Government's option, the wastes may be turned in to the MMAC hazardous waste program for disposal.
- 8.3. All uncontaminated, non-hazardous waste materials will be separated for removal and disposal in a state approved sanitary landfill.
- 8.4. The contractor will utilize good housekeeping to maintain a state of cleanliness at the scene of site operations and will complete site cleanup and restoration as appropriate at the completion of the job.

9. Testing:

- 9.1. The contractor will bear all costs for testing which may be called for in the several specifications sections that follow.

10. Cutting and Patching:

- 10.1. Definition: Includes cutting and patching of both previously existing work and nominally completed portions of contract work. Excludes shop fabrication of work, and normal installation procedures including the drilling of holes to install fasteners. Excludes special categories of work identified as alterations, demolition, excavating, grading, planting, cleaning, removal/replacement of noncomplying work and similar activities, although some of these activities may require cutting and patching.
- 10.2. Requirements for cutting and patching will be described by the contractor in its work plan or delivery order proposal. The contractor will describe the extent of cutting and patching required, how it is to be performed and why it cannot be avoided.
- 10.3. The contractor will:
 - 10.3.1. Not cut and patch structural elements in a manner that would reduce their load-carrying capacity or load-deflection ratio.
 - 10.3.2. Not cut and patch operating elements or safety related components in a manner that would result in reducing their capacity to perform as intended, or result in increased maintenance, or decreased operational life or safety.
 - 10.3.3. Use materials that are identical to existing materials. If identical materials are not available or cannot be used where exposed surfaces are involved, use compatible materials that match existing adjacent surfaces to the fullest extent possible regarding visual effect. Use materials whose installed performance will equal or surpass that of existing materials.
 - 10.3.4. Patch with durable seams that are as invisible as possible.

10.3.5. Restore exposed finishes of patched areas and extend finish restoration into retained adjoining construction in a manner that will eliminate evidence of patching and refinishing.

11. Cleanup:

11.1. All debris resulting from the proposed work will be removed from the premises and at the completion of the project, the area occupied and/or utilized during construction will be left in its original condition before construction began.

12. Stored Materials:

12.1. All materials stored at the site will be stored above ground and protected from the elements by being covered with a waterproof covering.

13. Performance Requirements for Completed Work:

13.1. In addition to the requirement that every element of the work comply with applicable requirements of contract documents, it is also required that the work as a whole (the project) be executed so as not to interfere in any way with the normal day-to-day occupancy and use of the existing facilities.

14. Warranties:

14.1. No special warranties are required. Standard manufacturers' warranties for installed equipment are acceptable.

15. Subcontractors:

15.1. If subcontractors are to be utilized, the prime contractor must provide the names and addresses of proposed subcontractors and the type and percentage of work they will provide to the CO. The prime contractor assumes responsibility for all services offered and produced. The prime contractor will be the sole source of contact regarding any task in this specification, including payment of any and all charges resulting from this contract.

15.2. The contractor will not subcontract all or any part of the work without first having obtained the written consent of the CO. Any such subcontract made or attempted to be made without such written consent will not be binding on and need not be recognized by the CO. Consent by the CO regarding subcontracting will not relieve the contractor of his responsibility for proper performance of all work covered in this specification.

16. Personnel Safety:

16.1. Personnel engaging in the investigation of hazardous waste or petroleum spill sites must be trained and qualified at a level commensurate with their job function and responsibilities, and to the degree of anticipated hazards. Anyone entering a site must be trained to recognize and understand the potential hazards to health and safety associated with sampling and/or cleanup of the site.

16.2. Personnel must be thoroughly familiar with the programs and procedures contained in the site Work Plan and Health and Safety Plan. The contractor will provide supportive evidence of the qualifications, experience and training of project personnel, including site workers, on-site management and supervisors, and health and safety staff.

16.3. Any visitors to the site must receive a safety briefing. Visitors are not permitted in the work area unless they have been trained, fit-tested and medically approved for respirator use, if site conditions

warrant the use of a respirator. The contractor is not required to provide the above, except for its employees.

17. Access to Flight Line:

- 17.1. Some of the sites requiring investigation are located inside the restricted air operations area (i.e. flight line) of Will Rogers International Airport. Contractor personnel entering this area must be escorted by an FAA employee or be properly credentialed. Contractor vehicles which enter the restricted area must carry \$1,000,000 of liability insurance and have proper permits for entry.

18. Handling of Drummed Waste:

- 18.1. The site Work Plan will provide Standard Operating Procedures for handling drums and containers, loading drums into overpack containers, and transporting containers to a containment area. All activities will follow procedures identified in the submitted plans regarding personnel protection and safety, air monitoring, established protocols, decontamination and handling procedures.

- 18.2. The contractor will define, in the site Work Plan, the practices and procedures for the safe handling of drums and other hazardous waste containers. In preparing these procedures, the contractor will consider the following:

18.2.1. OSHA, 29 CFR Parts 1910 and 1926 - This includes the general requirements and standards for storing, containing, and handling of chemicals and containers, and for maintaining equipment used for handling materials.

18.2.2. EPA, 40 CFR Part 265 - This stipulates requirements for types of containers, maintenance of containers and containment structures, and design and maintenance of storage areas.

18.2.3. DOT, 49 CFR Parts 171-178 - This stipulates requirements for containers and procedures for shipment of hazardous wastes.

18.2.4. MMAC waste storage and turn in procedures - To be furnished by the FAA Environmental and Safety Staff, AMP-100.

- 18.3. In the event of a spill while handling drums or other containers, the contractor will perform emergency spill response, including notification, containment and cleanup in accordance with the latest revision of Order AC 1050.4. To the fullest extent practicable, the contractor will provide for the prevention of such spills.

19. Documentation and Data Presentation:

- 19.1. The contractor will provide a copy of all documents generated under the terms of this contract, except for raw data produced by the laboratory during sample analysis. This includes, but is not limited to, such things as field notes, chain of custody forms, interview notes, and laboratory quality control/quality assurance data which may be handwritten, type written or computer generated. The contractor will keep sufficiently detailed records to substantiate and reconstruct the decision-making process for his portion of the agreement. All data will be adequate to establish the traceability of standards, instrumentation, samples, and environmental information. The contractor is not required to retain the material once it has been provided to the FAA.

- 19.2. The contractor will ensure that data generated under this contract is not disclosed to unauthorized personnel and data access is restricted to authorized personnel. The contractor will not present any data without the approval of the CO.

- 19.3. Sampling results will be included as an appendix to all final reports. These may be attached to the report or bound separately. These results will include the following:
- 19.3.1. A copy of the Sample Analysis Request Form that was submitted with the sample;
 - 19.3.2. A completed Chain of Custody form with the appropriate signatures;
 - 19.3.3. A tabulation of the analytical results along with the following:
 - 19.3.3.1. Date and time of sample collection,
 - 19.3.3.2. Sample identification number,
 - 19.3.3.3. Sample matrix,
 - 19.3.3.4. The parameters analyzed and their corresponding detected concentrations (When no concentration can be detected, the value to be reported will be "less than" the method detection limit.),
 - 19.3.3.5. Date and time of sample extraction,
 - 19.3.3.6. Date and time of analysis,
 - 19.3.3.7. The method detection limit for each parameter,
 - 19.3.3.8. A reporting of accuracy for each parameter,
 - 19.3.3.9. Reagent blank results, for each 8 hour shift,
 - 19.3.3.10. A reporting of precision for each parameter along with acceptability limits,
 - 19.3.3.11. Surrogate spike results along with acceptability limits,
 - 19.3.3.12. Citation of the analytical method used.
 - 19.3.4. Reporting of all Gas Chromatography (GC) and Gas Chromatography Mass Spectrometry (GC/MS) data may include, at the discretion of the COR:
 - 19.3.4.1. Quality Control (QC) matrix spike results,
 - 19.3.4.2. QC matrix spike acceptance criteria,
 - 19.3.4.3. Copies of the sample chromatograph with retention times appropriately labeled (Unknown consequential peaks (25% of internal standard) will be labeled "Unknown".),
 - 19.3.4.4. Copies of working standard chromatographs,
 - 19.3.4.5. Detailed written description of the extraction method employed per parameter grouping, and
 - 19.3.4.6. A short description of the method of analysis employed.

19.3.5. Reporting of all GC/MS data may include:

19.3.5.1. Copies for the reconstructed total ion chromatographs for each sample extract,

19.3.5.2. Copies of the mass spectra of compounds identified in the sample extract. Copies of the mass spectra of tentatively identified non-priority pollutant compounds.

20. Design Certification:

20.1 The design features of the work will be accomplished by or under the direct supervision of a registered professional engineer licensed to practice in the State of Oklahoma. The final documents will be stamped and signed by the engineer (See Division 2).

21. Miscellaneous and General Provisions:

21.1. The contractor will comply with all applicable codes and standards, including laws, governing regulations, health and safety codes and industry standards. Refer to individual specification sections for specific codes and standards.

22. Permits:

22.1. Where governing regulations and imposed codes and standards require notices, permits, licenses, inspections, tests and similar items or actions to lawfully proceed with the required work, the contractor will obtain those items and take those actions in accordance with the regulations of the governing authority. The costs of such permits, licenses, inspections, etc., are the obligation of the contractor.

SECTION 050 Contract Administration Data

1. Project Engineer/Manager:

1.1. The AE will appoint a project engineer (manager) for each project to serve as the single point of contact and liaison between the AE and the Government. Upon issuance of the Notice to Proceed for design of each project, the AE will advise the CO of the name of the individual so designated. The AE's project engineer/manager will be responsible for the complete coordination of all work required.

2. Records of Design Meetings/Conferences and Visits:

2.1. The contractor will provide a summary of each meeting attended, telephone conversation or other communication to the COR and regulatory agency representatives (if participating in the meeting or communication) within ten (10) days of the meeting or communication. The summary will give the date, location and purpose of the meeting or communication, people involved, items discussed, problems addressed and resolved and action items for follow up. This applies to any meeting or communication where significant decisions are made regarding field operations or any other aspect of work performed under this contract.

2.2. The contractor will hold project status meetings and submit monthly technical progress reports within ten (10) days after the end of each month to the COR.

3. Deliverables for technical and engineering support delivery orders will be addressed in each task order.

DIVISION 1 FIELD WORK PLANS

SECTION 110 Preinvestigation Activities

1. General:

1.1. Scope

- 1.1.1. This section describes pre-investigation activities necessary for the performance of investigative actions that may be performed under delivery order. These pre-investigative activities consist primarily of reviewing existing documentation relevant to the task and preparation of appropriate work, quality assurance, health and safety and other related plans.

2. Background Investigation:

- 2.1. The purpose of the background investigation is for the contractor to become familiar with information developed during previous investigations so that data gaps may be identified. Project documents such as work, safety and sampling plans which enable the safe, accurate collection of this data are then prepared and submitted prior to initiation of field work.
- 2.2. The contractor will review pertinent data and plans on file with FAA, EPA, and state and local agencies for the purpose of determining additional data needed to complete any site investigation.

3. Project documents:

- 3.1. Prior to the start of any field work the contractor will prepare and submit the project documents shown in paragraphs 3.3. through 3.7. for approval.
- 3.2. Project Schedule
 - 3.2.1. The contractor will provide a project schedule showing the initiation, coordination, interdependency and duration of work elements. The project schedule for long-term projects will include a phase/task bar chart or critical path diagram. The schedule will include as a minimum initiation and completion dates for the tasks shown below:
 - 3.2.1.1. Prepare Health and Safety Plan
 - 3.2.1.2. Prepare Field Sampling Plan
 - 3.2.1.3. Prepare Quality Assurance Plan
 - 3.2.1.4. Prepare Investigation Derived Waste Plan
 - 3.2.1.5. Prepare Work Plan
 - 3.2.1.6. Perform site work
 - 3.2.1.7. Submit draft report
 - 3.2.1.8. Submit final report

- 3.2.2. The contractor will submit each of the plans shown in paragraphs 3.3 through 3.7 for review and approval prior to initiating field work and should allow adequate time in his schedule for this review.
- 3.2.3. The contractor will notify the COR of any changes in the schedule. Scheduling changes must be reviewed and approved by FAA prior to initiation.
- 3.2.4. The contractor is responsible for scheduling on-site work at times when temperature conditions permit sampling and working outdoors.

3.3. Health and Safety Plan

- 3.3.1. A site-specific Health and Safety Plan (HASP) will be prepared for on-site personnel to assign responsibility, establish personnel protection standards, specify safe operating procedures, and provide contingencies that may arise during the execution of the work. Appropriate personnel protection measures should be defined for field activities at all FAA sites. The appropriate level of protection concerning on-site activities will be initially chosen based on assumed hazard/toxic conditions then re-established based on actual monitored and measured levels of contamination or field activities. The contractor will assume responsibility for personal protection and safety of personnel while working at MMAC and for updating the HASP as conditions warrant. The plan will address specific hazards which may be present at each site to be investigated.
- 3.3.2. The contractor will submit the site-specific HASP to the COR prior to commencing field activities. This plan must meet the requirements of 29 CFR 1910.120 and will include as a minimum the following elements as described in the referenced CFR:
 - 3.3.2.1. Site characterization and analysis (1910.120(c))
 - 3.3.2.2. Site control (1910.120(d))
 - 3.3.2.3. Training (1910.120(e)) - The contractor will provide certificates or other proof of proper training.
 - 3.3.2.4. Medical surveillance (1910.120(f))
 - 3.3.2.5. Engineering controls, work practices, and personal protective equipment for employee protection (1910.120(g))
 - 3.3.2.6. Monitoring (1910.120(h))
 - 3.3.2.7. Site safety and health plan (1910.120(i))
 - 3.3.2.8. Handling drums and containers (1910.120(j))
 - 3.3.2.9. Decontamination (1910.120(k))
 - 3.3.2.10. Emergency response (1910.120(l))
 - 3.3.2.11. Illumination (1910.120(m))
 - 3.3.2.12. Sanitation at temporary workplaces (1910.120(n))

3.3.3. The HASP must include the following:

3.3.3.1. Policy statements of lines of authority and accountability for plan implementation, plan objectives and roles of the site health and safety officer or manager and staff.

3.3.3.2. Procedures for identifying and controlling workplace hazards.

3.3.3.3. Procedures for communication to employees of various plans, work rules, standard operating procedures and practices that pertain to individual employees and supervisors.

3.3.3.4. Training supervisors and employees to develop the skills and knowledge needed to perform their work in a safe and healthful manner.

3.3.4. An Emergency Plan for on-site emergencies will be prepared which documents policies and procedures for responding to site emergencies. This plan will be prepared as a discrete section of the HASP and will:

3.3.4.1. Be compatible with the pollution response, disaster, fire and emergency plans of local, state and federal agencies.

3.3.4.2. Comply with the latest revisions of Order AC 1050.4, "Spill Prevention and Response Plan," and Order AC 1900.12 "MMAC Emergency Operations Plan."

3.3.5. The health and safety plan will otherwise comply with the applicable portions of the latest revisions of FAA Order 3900.19, "Occupational Safety and Health (OSH) Policy" and Aeronautical Center Order AC 3900.21, "Mike Monroney Aeronautical Center Occupational Safety and Health Program."

3.3.6. The HASP will be reviewed and updated in response to new or changing site conditions or information.

3.4. Field Sampling Plan

3.4.1. Development of reliable and quality soil, water, air and other environmental data is a critical element in any investigation. The site-specific Field Sampling Plan will include the following:

3.4.1.1. Identification of FAA MMAC site(s) to be investigated.

3.4.1.2. Locations of soil, sediment and surface water sampling points, areas to monitor for airborne contaminants, and monitoring/potable wells for measurement and sampling.

3.4.1.3. Justification for the locations and number of samples and monitoring points based on FAA or other available records.

3.4.1.4. Explanation on the type of data and information to be obtained and intentions for use regarding previous data verification, extent and levels of contamination and, if appropriate, targeted contaminants for control and removal. This includes the desired level of analytical accuracy and detectability of parameters to be tested and analytical methods to be used to achieve the desired level of accuracy.

- 3.4.1.5. Monitoring well design and construction details.
- 3.4.1.6. Sampling methods for each type of medium and sampling site. This includes sample marking, preservation and holding times.
- 3.4.1.7. Organization flow chart for sampling activities indicating names, titles and tasks to be performed. Give names and phone numbers for contractor and laboratory contacts.
- 3.4.1.8. Field apparatus to be used including make and model of field instruments and detectors.
- 3.4.1.9. A description of how contaminated investigation by-products will be handled and disposed of. By-products include such materials as decontamination solutions, disposable equipment, drilling muds and cuttings, well-development fluids, well-purging water, and spill-contaminated material.
- 3.4.2. Sampling will be conducted according to the most recent approved EPA, ASTM or other industry accepted standards.
- 3.4.3. Monitoring wells will be designed to comply with the Rules and Regulations of the Oklahoma Water Resources Board (OWRB) for construction of such wells.
- 3.4.4. Any changes to the sampling plan will be in writing or otherwise documented and agreed to by the COR.
- 3.4.5. The contractor will ensure that:
 - 3.4.5.1. Representative samples are collected which accurately portray the properties of the media being sampled.
 - 3.4.5.2. The degree of sampling and analytical accuracy and precision will be sufficient to estimate reliably the chemical characteristics of the sampled media to applicable regulatory thresholds and to attain an accuracy required by EPA, ODEQ, or other appropriate regulatory agency for the parameters to be tested.
- 3.4.6. Portable Gas Chromatography
 - 3.4.6.1. The contractor may choose to use a portable Gas Chromatograph (GC) at the site to analyze soil samples for chemical contamination and to screen samples for further testing.
 - 3.4.6.2. All analyses of samples for media destined for transportation, treatment or disposal must be performed in an offsite accredited laboratory.
 - 3.4.6.3. The contractor will notify the FAA of its intention to use a portable GC prior to field sampling. Use of a portable GC to identify ground material contamination and the success of excavation efforts, when applicable, is acceptable.
 - 3.4.6.4. Only qualified personnel trained in the operation of the specific model and brand of portable GC will be allowed to prepare and analyze samples and interpret analytical results. Sampling and analyses must follow QA/QC procedures described in the Quality Assurance Project Management Plan and as required for

the specific model and brand of GC. QA/QC data will be provided with the analytical results.

3.4.7. Abandoned or damaged monitoring wells must be plugged and otherwise sealed according to OWRB Rules and Regulations.

3.4.8. Specific requirements for soil and ground water sampling are described in Attachment B.

3.5. Quality Assurance Project Plan (QAPP)

3.5.1. To develop reliable monitoring data, the contractor will prepare a Quality Assurance Project Plan with respect to sampling, analysis, and data handling for all field and laboratory data to be generated that is specific to the sites in question.

3.5.2. The QAPP will contain, at a minimum, the guidelines presented in the EPA document QA/G-5 "Guidance for Quality Assurance Project Plans". Approval of the QAPP must be received prior to initiation of environmental measurements. This plan will include, but not be limited to, the following points, which are further elaborated in Attachments C and D:

3.5.2.1. Project quality assurance objectives and criteria for data measurements, in terms of precision, accuracy, and completeness for each parameter in each matrix (i.e. soil, water and air).

3.5.2.2. Plans to routinely assess the precision, accuracy, and completeness of data including specific procedures, practices and calculations.

3.5.2.3. A list of key individuals involved in the plan accompanied by their qualifications, functions and responsibilities.

3.5.2.4. A detailed description of sampling procedures, including a copy of the sampling manual (if applicable) and outlining cleaning procedures for all sampling equipment.

3.5.2.5. A description of sample shipment, receiving and documentation procedures (see Attachment C for requirements).

3.5.2.5. A description of field and laboratory chain of custody procedures. Include a copy of the Chain of Custody form to be used in the project. (see Attachment C).

3.5.2.6. A copy of the laboratory Quality Control Plan (QCP) will be submitted with the QAPP for review. The laboratory QCP will address the items shown in Attachment D to this SOW.

3.5.2.7. Describe performance and system auditing procedures for all measurement systems (field, laboratory, etc.) including degree and frequency of audits, and operational checks of field procedures (if applicable); frequency of Quality Assurance reports to management and corrective action procedures will be detailed.

3.5.2.8. All data will be reported in units of measurements consistent with similar data reported to the EPA, i.e., ug/l, ug/kg, etc. The reporting format to be used for the project will be included with submission for review and comment.

- 3.5.2.9. Data validation methodology will be described including predetermined acceptability limits for precision and accuracy and correction procedures when acceptability limits are exceeded.
- 3.5.2.10. Provide an outline of calibration procedures, references, and frequency for analytical and field instrumentation.
- 3.5.2.11. Provide preventive maintenance procedures and schedules for analytical and field instrumentation.
- 3.5.2.12. Changes to laboratory Standard Operating Procedures that occur during the contract will be submitted when they occur.
- 3.5.3. The contractor is responsible for all interactions with the laboratory and will ensure that the laboratory follows the QA/QC procedures as described in the laboratory QCP and as required by the EPA Contract Laboratory Program, if CLP or equivalent documentation is requested.
- 3.5.4. Analytical methods used will be EPA approved (NIOSH for air monitoring) standard methods unless otherwise approved by the CO/COR. Alternate methods must be properly validated and standardized methods for which precision, accuracy and other supportive information have been generated and must satisfy project requirements for precision, accuracy, specificity and sensitivity.
- 3.5.5. Quality control reports will be submitted to the COR weekly during field activities. The weekly report will contain at least the following: (1) location of work; (2) weather information; (3) work performed; (4) specific inspections performed and results; (5) problems identified; (6) corrective actions taken, if any; (7) verbal or written instructions from government personnel to perform retesting or additional testing; (8) type of tests performed, samples collected, personnel involved, and test results; (9) general remarks; (10) calibration procedures and recordings; and (11) contractor's certification.
- 3.5.6. It is the responsibility of the contractor to report to the COR in writing within two (2) working days any significant problems with analytical procedures, instrument calibrations or other QC problems along with corrective actions taken to resolve the problems. Reports of significant problems should be forwarded promptly to the QA laboratory.
- 3.5.7. The contractor will submit a quality control summary report at the end of the sampling/construction effort.
- 3.5.8. Performance audits will be conducted to assess quantitative data systematically and independently to determine if analytical results are within acceptable control limits.
- 3.5.8.1. The QAPP will describe the types and percentages of samples to be supplied to the QA laboratory, for external QA (performance audit) checks.
- 3.5.8.2. In support of performance audit checks, the QA laboratory may provide the COR with pre-cleaned sample containers and a sample handling protocol based on EPA guidelines for the external QA samples. The COR will ship these samples to the QA laboratory for analysis.

3.6. Investigation-Derived Waste Plan (IDW Plan)

- 3.6.1. The contractor will prepare a plan for managing investigation derived wastes such as soil cuttings, drilling muds, purged groundwater, decontamination fluids, disposable sampling equipment and disposable personal protective equipment. The contractor will consider the guidelines presented in EPA's OERR directive 9345.3-02, "Management of Investigation-Derived Wastes During Site Inspections" in developing this plan.

3.7. Work Plan

- 3.7.1. The purpose of the work plan is to efficiently schedule resources such as personnel, equipment and laboratory services. It should specify the administrative and logistic requirements of the project such as personnel requirements and responsibilities, equipment requirements, subcontractor requirements, special training requirements, schedule and budget and mobilization/demobilization procedures.
- 3.7.2. Elements of the work plan pertaining to information developed in other plans required by this contract may be incorporated by reference.

DIVISION 2 DESIGN REQUIREMENTS

SECTION 210 General

1. The AE will furnish all services, materials, supplies, personnel, and supervision required to fully investigate and design each project. The services required for each project will be negotiated under a separate statement of work and delivery order. See Section 010 Scope of Work.
2. Should the AE receive requests for work which he considers to be beyond the scope of this contract, the CO should be notified immediately. The AE should not start work on these items until given notice to proceed by the CO. An increase in construction costs over initial 100% of the budget program cost estimate will not be a basis for claim against the Government to increase the AE fee unless it has been caused by a change in the agreed scope of work.
3. The design features of the work will be accomplished by or under the direct supervision of a registered professional engineer in the particular field involved. The design features will be reviewed and approved by an engineer licensed to practice in the State of Oklahoma or other state where work may be performed. The final documents will be stamped and signed by an engineer registered in the particular field involved. The final documents will conform to the local, city and state construction permit requirements.
4. Reports prepared for submission to the Oklahoma Corporation Commission regarding the Underground Storage Tank program will be signed by an individual certified by the OCC for such submittals.

SECTION 220 Design Criteria

1. The extent of work will be identified in the Statement of Work for each project. The AE will prepare all designs in conformance with the applicable portions of the following:
 - 1.1. Department of Transportation (DOT) and Federal Aviation Administration (FAA) regulations required for design of each project. Current local, state, and national codes, regulations, and specifications involving architectural, civil, electrical, environmental, and mechanical disciplines associated with the design and construction of each project. This includes American Disability Act (ADA) and applicable safety and health standards issued by the Occupational Safety and Health Administration (OSHA).
 - 1.2. National Fire Codes as published by the National Fire Protection Association.
 - 1.3. All other DOT or FAA documents deemed applicable by the Contracting Officer or RE for each project. The Government will make these documents available to the AE as required.
2. The project statements of work will indicate design objectives. The AE is encouraged to suggest changes which would lower construction costs or improve performance but would not seriously deviate from the overall requirements.
3. The AE will design each project to provide a complete and usable facility within the given target construction cost provided by the Government. The target construction cost will include the construction contract, Government-furnished equipment, and contractor-furnished equipment.
4. The FAA's record drawing files will be made available to the AE. The AE is responsible for the reproduction of drawings and any associated costs.
5. All data furnished by the Government will remain the property of the Government and will be returned prior to or upon completion of the work and prior to final payment for services rendered.

6. For all references made on the drawings or in the specifications related to work, which is not included in contract, it will be responsibility of the AE to indicate specifically who is to perform or furnish that work. The AE will indicate the coordination required.

SECTION 230 Preparation of Project Manual (Type A)

1. The Project Manual or report will include all information, narratives, data, and computations necessary to support and justify the project developed and will be in such detail as to permit complete understanding of the project requirements. The Project Manual will be prepared in accordance with MMAC publication "MMAC Design Guidelines", which will be furnished to the AE.

SECTION 240 Preparation of Drawings (Type A and B)

1. All drawings will be computer generated, plotted in ink, and prepared by such methods and quality of workmanship to permit clear and legible reproduction including reproduction at one half scale.
2. A sample format of drawing sheet with title block, etc., will be furnished by the Government.
3. All drawings and other data submitted for review will be dated and marked to indicate stage of drawings.
4. AE will use building names, room numbers and room designations as provided by the FAA.
5. All drawings will be prepared in accordance with FAA publication "MMAC Design Guidelines", which will be furnished to the AE.

SECTION 250 Preparation of Specifications (Type B)

1. Specifications - The specifications will be complete and clear and will amplify all information shown on the drawings; explain methods of construction considered necessary; and specify the detailed requirements for materials and equipment. The specifications will be written in Construction Specification Institute (CSI) Division Format in a manner which will permit full and free competition among bidders. Specifications will be consistently organized and written using CSI subdivisions and paragraph numbers. Each sheet will be identified by section, page and project title. The specifying of a feature peculiar to one product will not be utilized unless necessary. Specification submittals will conform to the following:

- 1.1. Preliminary 35% Submittal - An outline listing all CSI specification section number and names required to be used in this project.
- 1.2. Regular 65% Submittal - A fully edited set of (marked-up) specifications applicable to this project.
- 1.3. Advanced-Final 100% Submittal - Completely edited set of specifications suitable for use as bid documents.
- 1.4. Final Submittal - Incorporate any revisions indicated on the advanced-final review comments and furnish letter quality bound specifications printed on 8-1/2" x 11" bond paper with 1" margin right and left, and 1-1/2" margin top and bottom, and editable electronic media of specifications.
- 1.5. The specifications for each construction project will contain the following statement:

"The construction contractor will furnish a copy of the construction drawings indicating in red any changes during the construction period from the original design. If there were no changes, the copy will be marked 'AS BUILT.' This copy of the construction drawings will be submitted to the COR within 30 days from project completion."

SECTION 260 Preparation of Cost Estimates (Type A and B)

1. Cost Estimates - Cost estimates will be prepared by computing quantities and applying prevailing local labor and material prices. The cost estimates will be conclusive and in detail to include all supporting items necessary for the accomplishment of a complete project. The first sheet of the cost estimate will summarize all costs by discipline. Standard estimating format will be used.
2. Preliminary and Regular Submittal - The cost estimate will be a control estimate based upon unit measures and quantities at current local price rates. This estimate should contain a minimum of 10% contingency.
3. Advanced-Final and Final Submittal - The cost estimate will be a complete and quantitative breakdown of all materials and items of work necessary to complete the project. All items of work will be listed separately and evaluated under their respective headings and in the required format for Government cost estimates. A reasonable percentage will be included for construction contractor's contingencies, profit, and overhead. Costs of the AE will not be included in the cost of the project.

SECTION 270 Preparation of Design Analysis, Code Reviews and Compliance (Type B)

1. The design analysis will include all information, narratives, data, and computations necessary to support and justify the design developed and will be in such detail as to permit complete understanding of the project design. The design analysis will be prepared in accordance with MMAC publication "MMAC Design Guidelines," which will be furnished to the AE.
2. The AE will submit construction documents to all appropriate local, city and state code enforcement officials for preliminary reviews. All costs associated with submitting for the preliminary review will be paid by the AE.
3. AE will provide a complete summary of code review comments, with an explanation of compliance with these comments.

SECTION 280 Preparation of Submittals, Review and Schedule

1. Electronic Media (Type A and B)
 - 1.1. The electronic media will include all submitted documents including design analysis, cost estimate, drawings and specifications and be provided in PDF and editable versions. Electronic media will be provided for every submittal phase.
 - 1.2. Drawings will be prepared in accordance with "MMAC Design Guidelines", which will be furnished to the AE.
 - 1.3. Specifications will be prepared using a word processing program compatible with the current Microsoft Word version being utilized by the government.
 - 1.4. At the conclusion of construction, if included in the statement of work, it will be the responsibility of the AE to provide corrected electronic media "AS BUILT" based upon the construction contractor's red-lined set.

- 1.5. Alternate submission formats may be acceptable provided the AE furnishes all documentation and materials required to make the alternate format acceptable and usable by the Government and receives written approval from the CO/COR.

2. Type A Submittals

- 2.1. Submittals will occur at the preliminary, regular, advanced-final, and final stages of a project. Smaller projects may have fewer submittals. The number of submittals and their due dates will be specified in the Statement of Work for each project. The Government review time for each submittal will also be indicated.
- 2.2. The AE will prepare and deliver submittals, with dated transmittal letter, to the RE.
- 2.3. Each submittal stage will incorporate the RE's review comments from the previous submittals.
- 2.4. Type A submittals will be in accordance with MMAC publication "MMAC Design Guidelines", which will be furnished to the AE.
- 2.5. To conserve paper, submittals will be printed double-sided to the fullest extent possible.
- 2.6. Drawings to be included would consist of, but not be limited to:
 - 6.1 Site plan with diagrammatic indications showing relationships of project components, topographical data, parking, sidewalks, site utilities, monitoring wells and sample locations.
 - 6.2 Building floor plans including space configurations, finishes, furniture, equipment, HVAC, plumbing, piping, telephone, power, and data requirements.
 - 6.3 Building elevations indicating materials, finishes and colors.
 - 6.4 Vertical building sections indicating relations of interior components.

3. Type B Submittals

- 3.1. Submittals will occur at the preliminary, regular, advanced-final, final, or as-built stages of the project.
- 3.2. Smaller projects may have fewer submittals. The number of submittals and their due date will be specified in the Statement of Work for each project. The Government review time for each submittal will also be indicated.
- 3.3. The AE will prepare and deliver submittals, with dated transmittal letter, to the RE.
- 3.4. Each submittal stage will incorporate the RE's review comments from the previous submittals.

3.5. The minimum submittal requirements are as follows:

3.5.1. Preliminary Submittal: This submittal will represent approximately 35% completion of the total project design and will include, but not be limited to:

3.5.1.1. Drawings:

- a. Site Plan showing building location, existing site conditions, topography, utilities and proposed site improvements.
- b. All Plan sheets identifying major demolition and construction items including asbestos, architectural, structural, roof, mechanical, and electrical sheets.
- c. Furnishings layout.
- d. Equipment layout and schedule.
- e. Elevations and Building Section.

3.5.1.2. Project Specifications

- a. Cover page and inside cover page with project name, FAA address, FAA work order number, submittal percentage, submittal date.
- b. Outline listing of the technical specifications to be used in this project.

3.5.1.3. Design analysis.

3.5.1.4. Cost estimate.

3.5.2. Regular 65% Submittal: The submittal will represent approximately 65% completion of the total design and will include:

3.5.2.1. All items from the Preliminary Design requirements.

3.5.2.2. Drawings:

- a. All items from the Preliminary Design requirements.
- b. Dimensioned and noted plan sheets.
- c. Enlarged detail plans.
- d. Construction sections, details, diagrams, schematics, interior elevations and dimensions.
- e. Schedules.

3.5.2.3. Project Specifications:

a. Preliminary specifications edited for this design application.

3.5.3. Advanced-Final Submittal: The submittal will represent 100% completion of the project design; fully coordinated, checked, stamped and signed. They will be complete to the extent that they could be used as contract bidding documents without further work. The following items will be included:

3.5.3.1. All items from previous submittals.

3.5.3.2. Advanced-final drawings fully noted and cross referenced, stamped and signed.

3.5.3.3. Advanced-final project specifications fully edited, stamped and signed.

3.5.3.4. Advanced-final cost estimate summary, itemized by CSI number for each discipline.

3.5.3.5. Advanced-final design calculations, economic analysis and support data.

3.5.3.6. Review code compliance 'work sheets' from all local, city and state code enforcement officials.

3.5.4. Final Design and Other Data Submittal: The AE makes corrections to advanced-final submittals and submits corrected originals of the final design drawings, specifications, cost estimates, and design data to the RE with a transmittal statement certifying the completeness, accuracy, and coordination of the documents. All Government furnished documents will be returned with the final submittal.

3.5.5. AS-BUILT Submittal: At the completion of construction the AE will, if included in the Statement of Work, perform such services as necessary to make corrections to the contract documents to reflect the actual construction. Such documents will be furnished within 60 days after construction completion.

4. Engineering Review Comments

4.1. Written review comments will be prepared and furnished by the RE to the AE at the conclusion of each submission for Government review. The AE will indicate concurrence or non-concurrence on each review comment. Should the AE not agree with the review comments, the AE's written dissension will be provided to the RE within 5 days and resolve at the submittal review conference. The AE will incorporate all review comments into the design documents prior to the next submittal.

4.2. Notwithstanding any review, approval and acceptance by the Government, the AE will be responsible for the professional and technical quality of all designs, drawings, specifications and other material produced under this contract; for the professional quality and adequacy of the service and material furnished; and for compliance with design criteria specified by the Government for use under this contract. The Government's review is made for the purpose of assuring the FAA that the functional requirements of the project and the facilities operation have been met. No responsibility for detailed design, systems function, specific proposals or omissions by the AE will be assumed by the Government. The Government's review is not intended to relieve the AE of their responsibility to meet contractual obligations.

5. Delivery / Quantity Schedules will be issued for each task order.

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DIVISION 3 COORDINATION AND COMMUNITY RELATIONS

SECTION 310 Project Coordination

1. General:

1.1. Scope

- 1.1.1. The nature of the work to be accomplished dictates that various meetings, conferences, and visits be conducted for the purpose of exchanging information and resolving problems relating to the work required under the contract. The contractor will provide effective project coordination/management and will be active with respect to community relations.

2. Project Coordination

- 2.1. The contractor will initiate and maintain the following activities to provide effective coordination:

- 2.1.1. Conduct joint project meetings with members of FAA, EPA, and the state to discuss activities underway and results/conclusions from completed task activities. The contractor will schedule and conduct a meeting prior to the start of work and interim meetings thereafter to allow for adequate input from the regulatory agencies.

- 2.1.2. The AE will attend and participate in all design conferences including orientation design and other meetings pertinent to the work to be accomplished. AE personnel representing each discipline to be involved in the conference material discussion will be present, unless prior approval is obtained from the RE. The AE will record minutes of all meetings and within five days furnish the RE with a typed written copy for confirmation. Confirmation notices should be numerically sequenced for ease of reference.

- 2.1.3. As a minimum, the AE will attend and participate in the following conferences/meeting:

- 2.1.3.1. Within 5 days (or as otherwise indicated in the task order) after the AE's receipt of the notice to proceed, a meeting will be held for orientation and discussion of the design criteria. The AE will have made a thorough study of the criteria submitted in the statement of work prior to this meeting. The primary purpose of this meeting is to answer questions and consider any AE proposed changes.

- 2.1.3.2. At the conclusion of the Government's review period for each submittal, a review conference will be held to discuss the review findings and ascertain the basis for developing the final data required.

- 2.1.4. Preparation of monthly technical progress reports detailing:

- 2.1.4.1. Site identification and activities underway,

- 2.1.4.2. Status of task items outlined in the SOW,

- 2.1.4.3. Updated Project Schedule for any Phase or delivery order in progress,

- 2.1.4.4. Percent of completion (e.g. in terms of man hours expended),

- 2.1.4.5. Problems or delays experienced during the reporting period,

- 2.1.4.6. Actions being taken to correct problems,
- 2.1.4.7. Activities projected over the next month,
- 2.1.4.8. Key personnel changes,
- 2.1.4.9. Additional work, justification and approval,
- 2.1.4.10. Billing to date.

- 2.2. The contractor will promptly respond to and comply with FAA MMAC's comments and directives involving all deliverables. Should questions arise due to comments from the regulatory agencies, the contractor will effectively meet with members of the FAA, EPA, and state to resolve the issue.
- 2.3. During the job, attendance will be required at joint project meetings with members of FAA, state, and EPA and at other meetings necessary to review the normal progression of the work in this SOW.

Section 320 Community Relations

- 1. Community relations activities will be conducted on an "as needed" basis. Inquiries regarding on-site remedial action from federal, state, or local officials, as well as citizens and the news media should be directed to the MMAC Public Affairs Staff. The contractor will also be prepared on an "as needed" basis to coordinate information regarding site activities with the Center's public affairs staff.

MMAC Site Cleanup Status
01/01/2026

SITE	KNOWN OR SUSPECTED PROBLEMS	STATUS
1. Tunnel System and System Training Building Sumps	Groundwater Contamination	Site Inspection completed 8/94. Site impacted by fuel and chlorinated organic plumes. Tunnel sump ventilation project to reduce vapors is complete. Actions proposed for Sites 4 and 18 will address this site also. NFRAP 10/24/02.
2. Outlet Lagoon / Lake Peachy	Surface water and Sediment Contamination	Site Inspection complete. Sediment found to contain low levels of polynuclear aromatic hydrocarbons (PAHs). "Analytical and Ecological Data Review for Lake Peachy" has been reviewed by ODEQ. Follow-up data analysis is complete. Final report has been accepted by ODEQ. NFRAP 10/29/99.
3. Radioactive Material Burial Site	Possible Ground-water Contamination	Site Inspection complete. Samples do not indicate impact to surrounding environment. Screening analysis report submitted to NRC for review. Site released for unrestricted use per NRC. NFRAP 8/18/00.
4. Fuel Transfer Station Area	Groundwater Contamination Spilled fuel	Site Inspection complete. OCC has approved risk-based corrective action plan for the site. Three-year monitoring plan is complete. Monitoring well decommissioning is complete. RTF sump fuel vapor ventilation project construction is complete. Case closed by OCC. NFRAP 10/27/99.
5. Army Air Corps Disposal Site	Soil Contamination. Land use must remain commercial/industrial.	Site Inspection complete. Soil contains elevated levels of PAHs. Risk-based corrective action plan accepted by ODEQ. NFRAP 2/13/98.
6. Cable Yard	NA	Site Inspection complete. NFRAP 8/15/94.
7. Steel Yard	NA	Site Inspection complete. Additional sampling showed that elevated lead levels did not originate from the site and could be naturally occurring. NFRAP 8/14/95.
8. Underground Fuel Tanks EM/ESS/STB Buildings	NA	Site Inspection complete. NFRAP 8/15/94.

9. Stoddard Solvent Tanks	NA	Site Inspection complete. NFRAP 8/15/94.
10. Surface Storage Tanks	Ground-water Contamination	Site Inspection complete. Site contaminated. Cleanup alternatives will be addressed with Site 4. Site 4 case closed by OCC. NFRAP 10/27/99.
11. Sanitary Sewer Releases	Possible Ground-water Contamination	Site Inspection complete. Smoke tests revealed leaks in 22 locations. Sewer pipe segments need to be repaired to prevent groundwater contamination. Repairs to known cross connection to storm sewer were completed on 1/29/01. OCAT reminded of other potential leaks identified in earlier smoke testing. Actions under way for site 18 will address this site as well.
12. Storm Sewer Releases	Possible Ground-water Contamination	Site Inspection complete. OPDES permit no. OK0043931 covering non-storm discharges became effective 3/1/01. Storm water pollution prevention plan was finalized and signed Sept. 14, 2001. NFRAP 10/9/01.
13. Base Maintenance	Subsurface product and possible groundwater contamination.	Site Inspection complete. Site contaminated with subsurface fuel. Free product recovery completed. OCC has approved risk-based corrective action plan for the site. Three-year monitoring program is complete. OCC has approved closure of this site. Monitoring well decommissioning is complete. Case closed by OCC. NFRAP 10/27/99.
14. Drum Staging Area	NA	Site Inspection complete. NFRAP 8/15/94.
15. CAMI Methylene Chloride Site	NA	Site Inspection complete. NFRAP 8/15/94.
16. Grounds Maintenance Building	NA	Site Inspection complete. NFRAP 12/94.
17. Building 50	NA	Site Inspection complete. NFRAP 8/15/94.

18. Chlorinated Organics Plume (COP)	Groundwater Contamination	The Chlorinated Organics Plume is located beneath land leased by FAA from the Oklahoma City Airport Trust (OCAT) for the Mike Monroney Aeronautical Center (MMAC). Land located at the southern portion of the plume is leased by the Federal Bureau of Prisons (Federal Transfer Center (FTC)) from OCAT. The plume of chlorinated organics (primarily solvents such as TCE), which is suspected to have resulted from leaks from waste oil and solvent storage tanks, the sanitary sewer, and/or the storm sewer systems, has contaminated the groundwater underlying MMAC. Groundwater concentrations of four pollutants were found to exceed Oklahoma Water Resources Board action levels. In 1996 and 1999, the Oklahoma Department of Environmental Quality (ODEQ) requested that FAA perform additional groundwater monitoring. In June 2002, a risk-based corrective action plan and closure request was submitted to ODEQ. However, in FY01 sampling indicated that the plume had moved under the FTC and FTC subsurface drainage pumps were discharging contaminated water. In October 2002, ODEQ denied MMAC's request for closure of this site and requested additional monitoring of off-site groundwater, as well as soil, indoor air, and surface discharge sampling at the FTC. In response, monitoring of groundwater and surface water has been conducted since FY95, and quarterly air monitoring has been conducted inside the FTC since early FY03. In FY05, a long range program plan was approved by ODEQ to address closure of the site. Also in FY05, the Screening Level Ecological Risk Assessment (SLERA) for Lake Peachy was completed and ODEQ concurred with MMAC's conclusions that there is no additional risk associated with contaminated discharges to Lake Peachy.
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18. Chlorinated Organics Plume (COP)-CONT	Groundwater Contamination	In FY06, semi-annual groundwater and surface water monitoring was conducted. In FY07, monitoring continued and air sampling was performed to evaluate the adequacy of FTC sump modifications. Based on the data collected, it appears that the sump modifications will be adequate to reduce concentrations of airborne contaminants to acceptable levels. FY08-12 testing and monitoring levels of the plume indicated no movement of the plume. FY-13 testing and monitoring indicated elevated levels of TCE in monitoring well 598 and therefore required new monitoring wells to be drilled to determine any further south movement of the plume. In 2015 – 2017, the FAA conducted expanded site characterization activities including willow and deep monitoring well installation, nuclear magnetic resonance geophysical logging, and soil vapor surveys. In 2020-2032, the FAA intends to build on the 2015-2017 site characterization work and complete a more extensive Supplemental Remedial Investigation that will close data gaps and update the Conceptual Site Model to include 3D visualization capabilities. The updates will enable the FAA to identify feasible remedial alternatives if appropriate and optimize the long-range COP program.
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Abbreviations:

NFRAP - No further remedial action planned

NRC - Nuclear Regulatory Commission

OCC - Oklahoma Corporation Commission

ODEQ - Oklahoma Department of Environmental Quality

ORBCA – Oklahoma Risk-Based Corrective Action

OWRB - Oklahoma Water Resources Board

PAH - polynuclear aromatic hydrocarbon

OPDES – Oklahoma Pollutant Discharge Elimination System

FTC –Federal Transfer Center, Federal Bureau of Prisons

Attachment B

Soil and Ground Water Sampling Requirements

1. Soil Sampling - All sampling points will be located in accordance with the Field Sampling Plan per field soil standard sampling methods ASTM D 1542/65, D 1586/67, D 3350/77 or equivalent. The contractor will:

- 1.1. Obtain relatively undisturbed soil samples using appropriate excavating equipment to dig sample pits, and, if necessary, use split spoon samples of two inches in outside diameter in case of contaminant extension. Continue digging or boring until reaching 2 feet below the water table or until contamination is no longer detected, whichever comes first, or other appropriate depth approved by the COR. The contractor will obtain the samples, using a split spoon to sample continuously from the surface to a depth of 2 feet below the water table. Soil sampling devices will be dedicated for each sample point or decontaminated in the field after each sample.
- 1.2. Adequate samples will be located to represent background conditions.
- 1.3. Record a complete description of each soil sample taken using a standard classification system. Also include physical characteristics, any presence of oils, viscous substances, color changes, ground water level, soil classification and thickness. Particle size analysis will be performed on representative samples to confirm field identification.
- 1.4. Provide pertinent soil sample information on well logging forms as appropriate.
- 1.5. Take representative samples and conduct laboratory analyses of EPA priority pollutants and organic non-priority pollutants as appropriate. (These are to be described in the Field Sampling Plan.)

2. Ground Water Investigation

- 2.1. All ground water sampling well locations will be based on the Field Sampling Plan. Monitoring wells will be required to accurately identify the extent of ground water contamination.
- 2.2. The contractor will be responsible for proper construction and subsequent sampling of any new wells. Well construction will comply with Oklahoma Water Resources Board Rules and Regulations for such construction. New wells and screens installed as part of this site investigation will be constructed of schedule 40 PVC.
- 2.3. Geophysical investigations will also be performed to define potential ground water contamination and to assist in locating monitoring wells. The contractor will meet with MMAC and state officials to discuss well installation procedures prior to the initiation of work.
- 2.4. The contractor will conduct slug or pump testing of monitoring wells as necessary to define aquifer characteristics.
- 2.5. The contractor will conduct ground water sampling on-site, and off-site as necessary to determine:
 - 2.5.1. Depth to water table which will be measured to 0.01 of a foot on a regular basis, tabulated and contoured.
 - 2.5.2. Levels of contamination laterally and vertically in the ground water (by concentration - ug/l or ppb).
 - 2.5.3. Contaminant behavior (stability, mobility, biological and chemical degradation).

2.5.4. Projected rate of contamination migration.

2.5.5. Direction and estimated flow rate in the aquifers determined to be contaminated or threatened both horizontally and vertically.

2.5.6. Degree and extent of ground water contamination in the underlying aquifers.

2.5.7. Geologic profile and cross-section to the depth of contamination as determined above.

2.5.8. A water-table contour map will be provided, as well as a map showing piezometric contours of any confined aquifers investigated. The direction(s) of ground water flow will be indicated on these maps.

2.5.9. Both vertical and horizontal permeability should be determined or estimated.

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Attachment C

Sample Shipment, Receiving and Documentation Requirements

1. Documentation Procedures:

- 1.1. All information pertinent to field activities, including sampling, must be recorded in various forms: logbooks, sample tags, photographs, etc. The contractor is required to provide proper documentation and document control of sampling and other on-site activities. Therefore, field workers are required to keep detailed records of inspections, investigations, photographs taken, and other pertinent information. All notes should be thoroughly reviewed before leaving the site. Photography must comply with the latest version of Order 1600.69, "FAA Facility Security Management Program".
- 1.2. Each sample must be sealed immediately after it is collected and labeled using waterproof ink. Labels may be filled out prior to collection to minimize handling of sample containers.
 - 1.2.1. Labels will include at least the following information: name of collector, date and time of collection, place of collection, sample number.
 - 1.2.2. Labels must be firmly affixed to the sample containers. Tags attached by strings or wires are acceptable when gummed labels are not available or applicable.
- 1.3. Sample identification numbers will consist of no more than 10 characters and will include a location designator, sample sequence number and other pertinent information. A description of the numbering system will be included in the QAPP.
- 1.4. All information pertinent to a field activity must be entered in a bound notebook with consecutively numbered pages. Entries in a log book must include at least the following: date and time of entry, purpose of sampling, name of field contact, description of sample, waste components and concentrations, number and size of samples taken, sample point description, date and time of sample collection, collector's sample identification number and/or name, site references, field observations and any field measurements taken.
 - 1.4.1. Notes should be as descriptive and inclusive as possible. Someone reading the entries should be able to reconstruct the sampling situation from the recorded information. Language must be objective, factual, and free of personal feelings or any other inappropriate terminology.
- 1.5. Photographs must be documented if they are intended to be used as a valid representation of an existing situation. For each photograph taken, the following must be recorded in the field logbook: date and time, signature of photographer, name and identification number of the site, general direction faced and description of the subject, location of site, sequential number of photograph and roll number.

2. Samples will be shipped to the analytical laboratory with sufficient time allowed for laboratory extraction of water samples within seven days of sample collection. Samples will be placed in clear plastic bags, labeled and placed in metal coolers according to EPA and contractor protocol. Vermiculite or similar material will be used as an absorbent and packing material. An original chain-of-custody form will be included inside the cooler, with a copy affixed to the outside. A custody seal will be used to ensure the cooler has not been opened and the samples were not tampered with prior to arrival at the laboratory.

3. The contractor will provide copies of all chain-of-custody forms and paperwork which follows the chain of possession and custody of any sample which may be offered for evidence, or which may form the basis of analytical test results used as evidence. Written procedures must be available and followed whenever evidence samples are collected, transferred, stored, or destroyed. The contractor will provide an accurate written record which can be used to trace possession and handling of a sample from the moment of its collection through analysis and its introduction as evidence.

3.1. The chain-of-custody form will include a record of sample numbers, number of containers shipped, project information, information on samples, method of shipment, and will be signed and dated.

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Attachment D

Laboratory Requirements

1. Quality Assurance/Quality Control Requirements - The primary laboratory and QA laboratory will provide a Quality Control Plan (QCP) for themselves, and any subcontract lab expected to perform any analyses under this contract. The QCP must demonstrate conformity with EPA recommended laboratory quality control procedures as described in such documents as 40 CFR 136 and EPA's "Handbook for Analytical Quality Control in Water and Wastewater Laboratories," September 1982. Each QCP will contain the following information:

- 1.1. Laboratory Organization and Responsibility - Indicate the personnel who will be involved in the analyses and what their responsibilities will be. Provide a brief resume for each person indicating his or her qualifications to perform the required work. The laboratory supervisor will have a B.S. in either Chemistry or Chemical Engineering and at least three years experience in analytical techniques needed to perform the tests listed in the Field Sampling Plan.
- 1.2. Quality Control Objectives for Measurement Data - Include a list of all parameters listed in the Field Sampling Plan, the reference of the method to be used (EPA or other) along with precision, accuracy and completion objectives as percent of samples analyzed for each parameter. The sensitivity and detection limit of each method must be indicated and must be sufficient for the purpose of the analysis. Describe interferences anticipated based on the method of analysis, matrix involved and other chemicals known to be present.
- 1.3. Sample Custody and Holding Procedures - Describe how EPA holding times will be met, sample preservative preparation methodology, sample container identification such as marking and labeling, field tracking forms to be used, name and title of designated sample custodian, and how sample custody is maintained until completion of analysis.
- 1.4. Calibration Procedures and Frequency - Describe frequency and method of calibration of all instruments and equipment. Describe preparation of reagent standards to be used for laboratory analyses.
- 1.5. Data Reduction, Validation and Reporting - Briefly describe procedures used to reduce, validate and report data generated in the laboratory.
- 1.6. Internal Quality Control Checks and Frequency - Describe how spike, duplicate, blank and standard analyses are used to maintain quality control, generation frequency and method of constructing quality control charts, participation in EPA or state laboratory quality control programs and laboratory management review of quality control results.
- 1.7. Performance and System Audits and Frequency - Describe any external performance audits or inspections made on the laboratory by EPA or state inspectors within the past year and the results of these inspections. Also describe management review frequency and procedures for evaluating analytical procedures, facilities and sampling methodologies. Include the type and frequency of quality assurance reports submitted to laboratory management. Also, describe actions taken if precision or accuracy data fall outside the boundaries of quality control chart limits.
- 1.8. Preventive Maintenance Procedures and Schedules - Describe preventive maintenance of laboratory equipment. Include such items as analytical balance, pH meter, water distillation unit.
- 1.9. Other information as required by the contractor.

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